

fableScalar and fable: a quintessence-like scalar and a 16-component spinor on the 4+4 pre-universe, their equations of state, and the Supernovae *Unite* CPL fits

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Abstract

We define two fields on the 4+4-dimensional pre-universe of P. L. Nash: **fableScalar**, a minimally coupled real scalar ϕ with self-interaction $V(\phi)$, and **fable**, a real 16-component $\text{Spin}(4, 4)$ spinor Ψ with self-interaction $V(s)$ of its scalar bilinear $s = \Psi^T \sigma_{16} \Psi$. For each we derive, and verify symbolically in Mathematica (358 of 358 assertions pass), the Lagrangian, the energy–momentum tensor, the field equation, the kinetic and potential energies, the energy density ρ , the pressure P and the equation of state $w = P/\rho$, and we integrate the resulting cosmologies with the pure-Rust translation of SUNDIALS 7.8.0 CVODE taken from the author’s rustSolveIt repositories. On the pre-universe the 8-volume element does not depend on the observer’s time, so there is no Hubble friction: a homogeneous **fableScalar** oscillates undamped with the virial average $\langle w \rangle = (n - 1)/(n + 1)$ for $V \propto \phi^{2n}$ (measured +0.0044 and +0.3349 over 100 time units for $n = 1, 2$); a static profile along the hidden coordinate is $w = -1$ on the observed sheet; and the bilinear of **fable** does not dilute at all. The null energy condition $\rho + P = 2 \text{ KE}$ holds identically for **fableScalar**, so $w_\phi \geq -1$ wherever $\rho_\phi > 0$. For **fable** on its homogeneous solutions $w = sV'(s)/V(s) - 1$: the author’s own mass term is exactly pressureless dust, and a potential whose slope changes sign carries w across -1 with no wrong-sign kinetic term. In the 4-dimensional reference models (a Friedmann equation on a separate FLRW frame) a thawing exponential **fableScalar** has the sign pattern of the Supernovae *Unite* CPL fit $(w_0, w_a) = (-0.861, -0.60)$ but never both numbers (closest, $\lambda = 1.4$: $(-0.674, -0.435)$), while a **fable** spinor with $V = V_0 + m s e^{-s/s_1}$ reproduces them to a few hundredths $((-0.851, -0.569)$, distance 0.032) with a genuine phantom phase in the past (minimum $w = -1.388$) and $\rho > 0$ throughout. CVODE, scipy’s DOP853 and Mathematica’s NDSolve agree to 1.3×10^{-10} (scalar) and 1.9×10^{-8} (spinor) in w . The framework has no gravitational sector, so which density gravitates, and the value of the scale-factor function a_4 , remain undetermined.

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1 Introduction

The task set by the author was to define a scalar field, `fableScalar`, and a 16-component spinor field, `fable`, each inspired by quintessence, each filling the pre-universe and evolving in time, each providing a physical mechanism for a time-varying equation of state of dark energy; to define, verify and test their Lagrangians and energy–momentum tensors, their kinetic and potential energies, pressures, energy densities and equations of state; to investigate the behaviour of w with numerical solutions and graphs; and to answer whether there is a connection to dark matter or dark energy.

The observational reference is the author’s note of 2026-09-15 (unpublished; its content is restated here). It uses the Chevallier–Polarski–Linder (CPL) parameterisation [1, 2]

$$w(a) = w_0 + w_a(1 - a), \quad (1)$$

and two fits to the Supernovae *Unite* data: forcing a constant equation of state gives $w = -0.764$, two standard deviations from -1 ; letting it vary gives $(w_0, w_a) = (-0.861, -0.60)$. The CPL line of the second fit reaches -0.861 today, crosses -1 at $a = 1 - 0.139/0.60 = 0.768$ and extrapolates to $w_0 + w_a = -1.461$ as $a \rightarrow 0$. The “phantom past” of that line lies at $a \lesssim 0.77$ and, below $a \approx 0.3$, outside the supernova range: it is an extrapolation of a straight line, not a measurement of $w < -1$. The note also recalls canonical quintessence [3, 4, 5], $L = \frac{1}{2}\partial\phi\partial\phi - V$, $\rho = \frac{1}{2}\dot{\phi}^2 + V$, $P = \frac{1}{2}\dot{\phi}^2 - V$, $\ddot{\phi} + 3H\dot{\phi} + V' = 0$; the distinction between thawing and freezing fields [6]; and the fact that a canonical scalar cannot cross the phantom divide $w = -1$ [7, 8].

Two kinds of statement are made in this paper and they are kept apart throughout. The first kind is *exact on the 8-dimensional pre-universe*: definitions, tensors, field equations and their

consequences, each proved symbolically in Part VI of the Mathematica notebook `claud-fable/claud-fable_Einstein-Rosen-2-Planes.nb` (Sections 23–25). The second kind belongs to *4-dimensional reference models*: a Friedmann equation on a separate FLRW frame, which the pre-universe does not supply (it has no gravitational sector and its scale-factor function a_4 is undetermined), used because it is the standard setting of the CPL fits. The reference models are never substituted into the pre-universe frame and never give a_4 a value.

Section 2 recalls the geometry; Sections 3 and 4 define the two fields and derive their properties; Section 5 states the numerical models; Section 6 the numerical method; Section 7 the verification; Section 8 the results; Section 9 the comparison with *Unite*; Section 10 the answers about dark matter and dark energy; Section 11 the limitations; Section 12 how to reproduce everything.

2 The pre-universe

The pre-universe has coordinates $X = \{x^0, \dots, x^7\}$, the flat metric

$$\eta = \text{diag}(+1, +1, +1, +1, -1, -1, -1, -1), \quad (2)$$

and the canonical frame (vielbein)

$$e = \text{diag}(\tan 6Hx^0, q, q, q, 1, p, p, p), \quad q = \frac{e^{-a_4(Hx^4)}}{\sin^{1/6} 6Hx^0}, \quad p = \frac{e^{+a_4(Hx^4)}}{\sin^{1/6} 6Hx^0}, \quad (3)$$

where H is the notebook's constant and a_4 an undetermined function. The curved metric is $g = e \eta e^T = \text{diag}(\tan^2 6Hx^0, q^2, q^2, q^2, -1, -p^2, -p^2, -p^2)$ and, for $0 < 6Hx^0 < \pi/2$,

$$\sqrt{g} \equiv \sqrt{\det g} = \sec 6Hx^0. \quad (4)$$

Two properties of this geometry drive everything below. First, the 8-volume element does not depend on x^4 : the observed 3-space (x^1, x^2, x^3) scales as q while the second sheet (x^5, x^6, x^7) scales as $p \propto 1/q$, so that $q^3 p^3 = 1/\sin 6Hx^0$; the volume element factorises as $\sqrt{g_8} = q^3 \cdot (\tan 6Hx^0 p^3)$, observed 3-volume times hidden 4-volume density. Second, on this frame the spin connection contributes nothing to the Dirac Lagrangian of a real 16-spinor (Part V of the notebook): $\gamma^\mu \Gamma_\mu^{\text{spin}}$ is a single-gamma term there, because the connection has no totally antisymmetric part. This is a property of the frame, not of the spinor.

Time, observer, energy components. The cosmological time is $t \equiv x^4$ ($g_{44} = -1$), the proper time of the geodesic observer $u = \partial_4$. For that observer, with the Hilbert tensor,

$$\rho \equiv T_{44}, \quad P_i \equiv T^i_i \quad (i = 1, 2, 3, \text{ no sum}), \quad w \equiv P/\rho \quad (\rho > 0). \quad (5)$$

Both fields are taken homogeneous on the observed sheet, so $P_1 = P_2 = P_3 \equiv P$.

The kinematic identification. The x^1 – x^3 frame entry at fixed x^0 plays the role of the observed scale factor:

$$a(t) \equiv q|_{x^0} \propto e^{-a_4(Ht)}, \quad \ln a = -a_4(Ht) + \text{const}, \quad H_{\text{obs}} \equiv \frac{\dot{a}}{a} = -H a'_4(Ht), \quad (6)$$

while the second sheet contracts at exactly the opposite rate, $\partial_4 \ln p = -H_{\text{obs}}$. This is a kinematic identification only. The observed sheet expands ($H_{\text{obs}} > 0$) exactly when $a'_4 < 0$, which is assumed and stated as a hypothesis, never derived; a_4 itself is never given a value.

3 fableScalar: the scalar field

3.1 Lagrangian, energy–momentum tensor, field equation

fableScalar is a real scalar $\phi(x)$ on the 8-manifold, minimally coupled, with self-interaction potential $V(\phi)$:

$$\mathcal{L}_\phi = \sqrt{g} \hat{L}_\phi, \quad \hat{L}_\phi = -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi). \quad (7)$$

The sign makes the x^4 kinetic term positive: $-\frac{1}{2} g^{44} \dot{\phi}^2 = +\frac{1}{2} \dot{\phi}^2$. The Hilbert tensor $T_{\mu\nu} = -(2/\sqrt{g}) \delta S / \delta g^{\mu\nu}$ is

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi + g_{\mu\nu} \hat{L}_\phi, \quad (8)$$

and the field equation is $\square \phi = V'(\phi)$ with $\square = (1/\sqrt{g}) \partial_\mu (\sqrt{g} g^{\mu\nu} \partial_\nu)$, which on the pre-universe reads

$$-\ddot{\phi} + \cos 6Hx^0 \partial_0 (\sec 6Hx^0 \cot^2 6Hx^0 \partial_0 \phi) + \frac{1}{q^2} \sum_{i=1}^3 \partial_i^2 \phi - \frac{1}{p^2} \sum_{i=5}^7 \partial_i^2 \phi = V'(\phi). \quad (9)$$

Part VI proves that $T_{\mu\nu}$ is symmetric; that the Hilbert prescription reproduces (8); the *off-shell* identity

$$\nabla^\mu T_{\mu\nu} = (\square \phi - V'(\phi)) \partial_\nu \phi \quad (10)$$

for a generic $\phi(x^0, \dots, x^7)$, so conservation on shell is exact and needs no assumption (with a control: off shell, on $\phi = x^0 + (x^4)^2$ with $V = \phi^2$, the divergence does not vanish); and the trace $T^\mu{}_\mu = -3(\partial\phi)^2 - 8V$.

3.2 Energies, density, pressure, equation of state

For a field of x^0, x^4 and the second-sheet coordinates define

$$\text{KE} = \frac{1}{2} \dot{\phi}^2, \quad \text{PE} = V(\phi), \quad G_0 = \frac{1}{2} \cot^2(6Hx^0) (\partial_0 \phi)^2, \quad G_5 = \frac{1}{2p^2} \sum_{i=5}^7 (\partial_i \phi)^2, \quad (11)$$

the kinetic energy along x^4 , the potential energy, the gradient energy along the hidden coordinate x^0 and the gradient energy along the second, timelike, sheet. Then

$$\rho_\phi = \text{KE} + \text{PE} + G_0 - G_5, \quad P_\phi = \text{KE} - \text{PE} - G_0 + G_5, \quad w_\phi \equiv w_{\text{Scalar}} = P_\phi / \rho_\phi, \quad (12)$$

and along x^0 itself $P_0 = \text{KE} - \text{PE} + G_0 + G_5$. The usual quintessence formulas are the case $G_0 = G_5 = 0$. Consequences, each an assertion of Part VI:

(S1) **The null energy condition holds identically:** $\rho_\phi + P_\phi = 2 \text{KE} \geq 0$ for every $\phi(x^0, x^4, x^5)$ and every V , hence $w_\phi + 1 = 2 \text{KE} / \rho_\phi$ and $w_\phi \geq -1$ wherever $\rho_\phi > 0$. Positivity is guaranteed for $V \geq 0$ and a field independent of x^5, x^6, x^7 . The gradients along the timelike second sheet enter ρ_ϕ negatively ($\phi(x^5)$ with $V = 0$ has $\rho = -G_5 < 0$); that sector is excluded from every statement about w . fableScalar cannot cross the phantom divide with positive energy.

(S2) **No Hubble friction in x^4 .** For $\phi = \phi(x^4)$ the field equation is $\ddot{\phi} = -V'(\phi)$, because $\sqrt{g} g^{44} = -\sec 6Hx^0$ does not depend on x^4 . Then $\rho_\phi = \text{KE} + V$ is conserved on shell. (Control: a friction term added by hand, $\ddot{\phi} = -3H_f \dot{\phi} - V'$, gives $\dot{\rho} = -3H_f(\rho + P)$, the Friedmann continuity equation, which is absent here.) The field oscillates undamped; on the exact solution $\phi = A \cos mx^4$ of $V = \frac{1}{2} m^2 \phi^2$, $w = -\cos 2mx^4$ with $\rho = \frac{1}{2} A^2 m^2$ constant, and the average of w over a period is 0: coherent oscillations in a quadratic potential are dust, the case $n = 1$ of the virial average $\langle w \rangle = (n - 1)/(n + 1)$ for $V \propto \phi^{2n}$ [13].

- (S3) **A static x^0 profile is a cosmological constant on the observed sheet.** With $V = 0$, $\partial_0(\sec \cot^2 \partial_0 \phi) = 0$ is solved by $\partial_0 \phi = C \sin^2(6Hx^0) / \cos(6Hx^0)$ for every constant C ; on it $G_0 = \frac{1}{2}C^2 \sin^2 6Hx^0$, independent of x^4 and of x^1, x^2, x^3 , and $\rho = G_0$, $P_1 = -G_0$: $w = -1$ exactly for the observer at fixed x^0 . Along the hidden direction, however, $P_0 = +G_0 = +\rho$: in eight dimensions this is an anisotropic stress, not a Λ term.
- (S4) **The volume bookkeeping.** Integrating the 8-dimensional action over the hidden coordinates gives $S_4 = \int d^4x a^3 \mathcal{V}_{\text{hid}} \hat{L}$ with the hidden 4-volume density $\mathcal{V}_{\text{hid}} \propto \tan(6Hx^0) p^3 \propto e^{3a^4} = a^{-3}$ (defined up to the formal integral over the three non-compact timelike directions). The reduced density $\rho_4 = \mathcal{V}_{\text{hid}} \rho_8$ is not separately conserved: $\dot{\rho}_4 + 3H_{\text{obs}}(\rho_4 + P_4) = 3H_{\text{obs}}P_4$, i.e. $\dot{\rho}_4 = -3H_{\text{obs}}\rho_4$ for *every* w . The a^{-3} is a volume effect that a cosmological constant would share; it is not a dark-matter signature. The locally measured density $T_{44} = \rho_8$ of $\phi(x^4)$ is exactly constant.

3.3 The 4-dimensional reference model

The standard quintessence system,

$$\ddot{\phi} + 3H_{\text{obs}}\dot{\phi} + V'(\phi) = 0, \quad H_{\text{obs}}^2 = \frac{1}{3}(\rho_m + \rho_r + \frac{1}{2}\dot{\phi}^2 + V) \quad (8\pi G = 1), \quad (13)$$

is what one obtains when the hidden volume is held fixed and the field sources the expansion on a separate FLRW frame. It is the reference model of the CPL fits and it is what Models A and A' of Section 5 integrate.

4 fable: the 16-component spinor field

4.1 Definition and fidelity to the original model

fable is a real 16-component spinor $\Psi(x)$ on the 8-manifold transforming under $\text{Spin}(4, 4)$ exactly as the model's Ψ_{16} of Section 11 of the notebook, with the notebook's Dirac matrices $T_{16}[a]$ ($a = 0, \dots, 7$, $\frac{1}{2}\{T_{16}[a], T_{16}[b]\} = \eta_{ab}\mathbf{1}_{16}$), the spinor metric σ_{16} (symmetric, with $\sigma_{16}T_{16}[a]$ antisymmetric), the curved Dirac matrices $\gamma^\mu = E_a^\mu T_{16}[a]$ built with the inverse frame, and the scalar bilinear

$$s \equiv \Psi^T \sigma_{16} \Psi. \quad (14)$$

With a self-interaction potential $V(s)$,

$$\mathcal{L}_\Psi = \sqrt{g} \hat{L}_\Psi, \quad \hat{L}_\Psi = \frac{1}{H} \Psi^T \sigma_{16} \gamma^\mu D_\mu \Psi - V(s), \quad D_\mu = \partial_\mu + \Gamma_\mu^{\text{spin}}. \quad (15)$$

On the canonical frame the connection term vanishes identically in $\hat{L}_\Psi$, so there

$$\hat{L}_\Psi = \frac{1}{H} \Psi^T \sigma_{16} \gamma^\mu \partial_\mu \Psi - V(s). \quad (16)$$

The author's own Lagrangian **La** of Section 11 is the case $V(s) = -(2M/H)s$ on the flat frame without volume factor: Part VI asserts that the Lagrangian helper *is* **La** there, and that its Euler–Lagrange equations *are* the sixteen equations **eLa** of Section 12. This is the fidelity anchor of the construction.

4.2 Field equation

Varying Ψ (kinetic matrix $A^\mu \equiv \sigma_{16} \gamma^\mu$ antisymmetric, mass matrix σ_{16} symmetric), Part VI finds by explicit variation that the Euler–Lagrange expression of $\sqrt{g} \hat{L}_\Psi$ is $\sqrt{g} [(2/H)(A^\mu \partial_\mu \Psi + \frac{1}{2} \mathcal{D} \Psi) - 2V'(s) \sigma_{16} \Psi]$ with $\mathcal{D} = \sigma_{16}(1/\sqrt{g}) \partial_\mu (\sqrt{g} \gamma^\mu)$, i.e. the field equation

$$\gamma^\mu \partial_\mu \Psi + \frac{1}{2} \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu) \Psi = H V'(s) \Psi. \quad (17)$$

In general $(1/\sqrt{g})\partial_\mu(\sqrt{g}\gamma^\mu) = [\gamma^\mu, \Gamma_\mu^{\text{spin}}]$ (covariant constancy of γ^μ). On the canonical frame the anticommutator $\{\gamma^\mu, \Gamma_\mu^{\text{spin}}\}$ vanishes (the stated hypothesis, asserted), so the second term of (17) is $\gamma^\mu \Gamma_\mu^{\text{spin}} \Psi = -3H \cot^2(6Hx^0) T_{16}[0] \Psi$ and (17) is exactly the Levi-Civita covariant Dirac equation $\gamma^\mu D_\mu \Psi = H V'(s) \Psi$. The divergence term does not vanish on an x^0 -independent spinor (witnessed on a constant spinor): *a spinor homogeneous in every coordinate but x^4 is not a solution*. The rescaling $\Psi = \sqrt{\sin 6Hx^0} \Psi'$ removes the term exactly for a generic $\Psi'(x^0, x^4)$: $\gamma^\mu \partial_\mu \Psi' = H V'(s) \Psi'$ with $s = \sin(6Hx^0) \Psi'^T \sigma_{16} \Psi'$. Since $(\gamma^4)^2 = -\mathbf{1}_{16}$, a $\Psi'(x^4)$ obeys $\partial_4 \Psi' = -H V'(s) \gamma^4 \Psi'$; it is an exact solution only when V' is constant (the mass and **lambda-mass** terms). For a nonlinear V the reduced equation still depends on x^0 through $V'(\sin(6Hx^0) s')$ (witnessed with $V = s^2$), and the exact problem is the (x^0, x^4) system.

4.3 Energy-momentum tensor

The frame variation of the covariant action gives, after symmetrisation (the variation of the spin connection produces a term antisymmetric in $\mu\nu$ that drops out),

$$T_{\mu\nu} = -\frac{1}{2H} (\Psi^T \sigma_{16} \gamma_\mu D_\nu \Psi + \Psi^T \sigma_{16} \gamma_\nu D_\mu \Psi) + g_{\mu\nu} \hat{L}_\Psi, \quad \gamma_\mu = g_{\mu\nu} \gamma^\nu. \quad (18)$$

Part VI proves it symmetric and *conserved on shell in all eight components* for a generic $\Psi(x^0, x^4)$, by substituting the field equation for every x^4 -derivative of the field. The tensor built with ∂_ν instead of D_ν has the same diagonal (so the same ρ and P) but differs off the diagonal (witnessed on the (x^3, x^4) component) and is not the conserved one.

4.4 Energies, density, pressure, equation of state

Define the kinetic bilinears along the observer's time and along the hidden coordinate, and the potential energy,

$$K_\Psi \equiv \frac{1}{H} \Psi^T \sigma_{16} \gamma^4 \partial_4 \Psi, \quad K_h \equiv -\frac{1}{H} \Psi^T \sigma_{16} \gamma^0 \partial_0 \Psi, \quad U_\Psi \equiv V(s). \quad (19)$$

For a field homogeneous on the observed sheet, on shell,

$$\begin{aligned} \hat{L}_\Psi &= sV'(s) - V(s), & P_\Psi &= sV'(s) - V(s), \\ \rho_\Psi &= V(s) + K_h, & K_\Psi &= sV'(s) + K_h, \end{aligned} \quad (20)$$

with $P_1 = P_2 = P_3$. $K_h = 0$ for $\Psi = \sqrt{\sin 6Hx^0} \Psi'(x^4)$ (because $\partial_0 \Psi$ is then proportional to Ψ and $\Psi^T \sigma_{16} \gamma^0 \Psi = 0$), and for those solutions

$$\rho_\Psi = V(s), \quad w_\Psi \equiv w = \frac{sV'(s)}{V(s)} - 1, \quad \rho_\Psi > 0 \iff V(s) > 0. \quad (21)$$

Consequences, each an assertion of Part VI:

- (F1) **The author's mass term is dust.** For $V = -(2M/H)s$, on shell $P_1 = 0$ identically and $\rho = -(2M/H)s$ (positive on the branch $Ms < 0$): $w = 0$ exactly. The spinor of the original model, promoted to a cosmological field, is pressureless cold dark matter.
- (F2) **w_Ψ is not bounded below by -1 .** $V = \lambda s^n$ gives $w = n - 1$: dust for $n = 1$, accelerating for $n < 2/3$, phantom for $n < 0$; $n = 0.236$ gives exactly Unite's constant $w = -0.764$. Since $w + 1 = sV'/V$, w crosses -1 where V' changes sign, and if $V > 0$ there the phantom divide is crossed with no wrong-sign kinetic term: the classical mechanism of the spinor quintom [10, 9]. The stability of perturbations is not examined here.

(F3) **No dilution on the pre-universe (Model E).** On shell, for a generic $\Psi(x^0, x^4)$, $\partial_4 s = 2\Psi^T \sigma_{16} \gamma^4 \gamma^0 \partial_0 \Psi$ (the other terms drop because $\sigma_{16} \gamma^4$ and $\sigma_{16} \gamma^4 \gamma^0$ are antisymmetric). That bilinear is not identically zero (witnessed on $\Psi = e_1 + x^0 e_2$), but it vanishes for $\Psi = \sqrt{\sin 6Hx^0} \Psi'(x^4)$, and from the reduced equation $\partial_4 s' = 0$ because $\Psi'^T \sigma_{16} \gamma^4 \Psi' = 0$. The bilinear, and with it V , ρ_Ψ and w_Ψ , is constant along the observer's time: no dilution, no oscillation, and a fixed equation of state set once by where s sits on the potential.

(F4) **The 4-dimensional reference model (Model B).** The reference model lives on a separate FLRW frame, never substituted into the canonical frame,

$$e_{\text{FLRW}} = \text{diag}(1, a(t), a(t), 1, 1, 1, 1).$$

On it Part VI finds $\gamma^\mu \Gamma_\mu^{\text{spin}} = \frac{3}{2}(\dot{a}/a)\gamma^4$, the Hubble term of the FLRW Dirac equation, and $d(a^3 s)/dt = 0$ on shell for every V : $s = s_0 a^{-3}$, the dust dilution law. (Control: without the Hubble term $d(a^3 s)/dt = 3(\dot{a}/a)a^3 s$, i.e. the pre-universe law $\dot{s} = 0$; the dilution *is* the Hubble term.) Then $w_\Psi(a)$ is the algebraic function (21) on $s = s_0 a^{-3}$ and the whole history is fixed by V .

4.5 The potentials of the numerics

Table 1 lists the six potentials of **fable** used in the numerics, with their closed-form equations of state (all asserted in Part VI); Table 2 the seven of **fableScalar**.

Table 1: The potentials of **fable** ($m, \lambda, V_0, \mu > 0$) and their equations of state $w = sV'/V - 1$. With $s = s_0 a^{-3}$ a crossing at $s = s_1$ happens at $a_c = (s_1/s_0)^{-1/3}$.

name	$V(s)$	$w_\Psi(s)$	behaviour on $s = s_0 a^{-3}$
mass	ms	0	dust at all times (the author's model)
lambda-mass	$V_0 + ms$	$-V_0/(V_0 + ms)$	dust early, -1 late; V_0 is a bare cosmological constant: Λ CDM with spinor dust
power	$ms + \lambda s^n$	$\frac{ms + n\lambda s^n}{ms + \lambda s^n} - 1$	for $n < 1$: dust $\rightarrow n - 1$ (accelerating for $n < 2/3$)
hilltop	$V_0 - \mu(s - s_*)^2$	$-\frac{2\mu s(s - s_*)}{V} - 1$	crosses -1 at s_* ; unbounded below, valid only for $s < s_* + \sqrt{V_0/\mu}$
lorentz	$V_0 + \frac{ms}{1+u}, u = \frac{s^2}{s_1^2}$	$\frac{ms(1-u)}{(1+u)^2 V} - 1$	positive everywhere; phantom for $s > s_1$, crosses -1 at s_1 , $w \rightarrow -1$ as $s \rightarrow 0$ and $s \rightarrow \infty$
expdamp	$V_0 + m s e^{-s/s_1}$	$\frac{m s e^{-s/s_1}(1 - s/s_1)}{V} - 1$	bounded below by V_0 ; same crossing pattern at s_1

Table 2: The potentials of **fableScalar** and the frozen initial values ϕ_i of the FLRW runs ($\dot{\phi}_i = 0$). The overall scale of V is fixed by the normalisation of Section 5.

name	$V(\phi)$	default ϕ_i	remark
exp	$V_0 e^{-\lambda \phi}$	0	thaws from rest; scaling attractor for $\lambda^2 > 3$ [11]
invpower	$V_0 \phi^{-\alpha}$	0.2	Ratra-Peebles; tracker $w = -2/(\alpha + 2)$ in the matter era
pngb	$V_0(1 + \cos \phi/f)$	$0.5f$	pseudo-Nambu-Goldstone boson [12]
quadratic	$\frac{1}{2}m^2 \phi^2$	1	oscillating, $\langle w \rangle = 0$
hilltop	$V_0(1 - \phi^2/\mu^2)$	0.1μ	used where $V > 0$
const	V_0	0	control: $w = -1$
quartic	$\frac{1}{4}\lambda \phi^4$	1	oscillating, $\langle w \rangle = 1/3$

5 The numerical models

Units and the Friedmann equation. $M_{\text{pl}}^2 = 1/(8\pi G) = 1$, $H_0 = 1$ (time in units of $1/H_0$), $a_0 = 1$, so $\rho_{\text{crit},0} = 3$; $\Omega_{m0} = 0.3$, $\Omega_{r0} = 8.4 \times 10^{-5}$, $\rho_m = 3\Omega_{m0}e^{-3N}$, $\rho_r = 3\Omega_{r0}e^{-4N}$; the independent variable of the FLRW models is $N = \ln a$, and

$$H^2 = \frac{\rho_m + \rho_r + \rho_{\text{field}}}{3} = \Omega_{m0}e^{-3N} + \Omega_{r0}e^{-4N} + \frac{\rho_{\text{field}}}{3}, \quad (22)$$

so $H(N=0) = 1$ exactly when the field carries $\Omega = 1 - \Omega_{m0} - \Omega_{r0} = 0.699916$ today. Cosmic time is carried as a state variable, $dt/dN = 1/H$, from N_0 ; the age at N_0 is added analytically as $t_{\text{age}}(N_0) = 1/(2H(N_0))$, the age of a radiation-dominated universe with that Hubble rate ($2.2 \times 10^{-5}/H_0$ at $N_0 = -7$). The deceleration parameter is $q_{\text{dec}} = -1 - (dH/dN)/H$ with $dH/dN = -(3\rho_m + 4\rho_r + 3(\rho + P)_{\text{field}})/(6H)$.

Initial conditions. Every FLRW run starts at $N_0 = -7$ ($a = 9.12 \times 10^{-4}$, $z = 1095.6$; radiation-matter equality is at $a_{\text{eq}} = \Omega_{r0}/\Omega_{m0} = 2.8 \times 10^{-4}$, and radiation still carries $\Omega_r = 0.2349$ at the start of every run in which the field is then negligible). The scalar starts frozen, $\dot{\phi}(N_0) = 0$, at the ϕ_i of Table 2. Whether a run thaws or freezes is read off the sign of dw/dN over the fit range $0.3 \leq a \leq 1$, not assumed from the potential's name: *thawing* if $dw/dN > -10^{-6}$ throughout, *freezing* if $< -10^{-6}$ throughout, *constant* if $|dw/dN| < 10^{-6}$, *mixed* otherwise.

Model A (fableScalar, self-consistent FLRW). State $y = (\phi, \dot{\phi}, t)$:

$$\begin{aligned} \frac{d\phi}{dN} &= \frac{\dot{\phi}}{H}, & \frac{d\dot{\phi}}{dN} &= -3\dot{\phi} - \frac{V'(\phi)}{H}, & \frac{dt}{dN} &= \frac{1}{H}, \\ H^2 &= \Omega_{m0}e^{-3N} + \Omega_{r0}e^{-4N} + \frac{\frac{1}{2}\dot{\phi}^2 + V(\phi)}{3}. \end{aligned} \quad (23)$$

Rescaling $V \rightarrow kV$ is not a symmetry of (23), so the scale is found by *shooting*: bisection on $\log_{10} k$ in $[10^{-6}, 10^6]$, 80 iterations or $\Delta \log_{10} k < 10^{-13}$, until $\Omega_{\phi}(a=1) = 0.699916$; the solver refuses (exit code 1) if the target is not bracketed or if $\rho_{\phi} \leq 0$ anywhere.

Model A' (fableScalar as a test field on the Unite CPL background). The same equations with $H^2 = \Omega_{m0}a^{-3} + \Omega_{r0}a^{-4} + \Omega_{DE0}f(a)$, $f(a) = \exp(3 \int_a^1 (1 + w(a')) da'/a') = a^{-3(1+w_0+w_a)}e^{-3w_a(1-a)}$, $(w_0, w_a) = (-0.861, -0.60)$. The field does not source H ; its scale is fixed so that $\rho_{\phi}(a=1)/3 = \Omega_{DE0}$.

Model B (fable, self-consistent FLRW). State $y = (s, t)$:

$$\frac{ds}{dN} = -3s, \quad \frac{dt}{dN} = \frac{1}{H}, \quad H^2 = \Omega_{m0}e^{-3N} + \Omega_{r0}e^{-4N} + \frac{V(s)}{3}, \quad (24)$$

with $s = 1$ today ($s_i = e^{21} = 1.3188 \times 10^9$ at $N_0 = -7$). The first equation has the closed solution $s = e^{-3N}$; integrating it numerically is deliberate, since it is the integrator that is being tested. Normalisation: $V \rightarrow kV$ with $V(1) = 3 \times 0.699916$, which leaves $w(s)$ unchanged exactly. The solver refuses if $V(s) \leq 0$ at the start or at any output point.

Model C (fableScalar on the pre-universe, homogeneous). $\ddot{\phi} = -V'(\phi)$ in x^4 (state $(\phi, \dot{\phi})$, $H = 1$); reported: KE, PE, ρ , P , w , the running trapezoidal average of w and the drift $\rho/\rho(0) - 1$, which must be at solver precision.

Model D (fableScalar on the pre-universe with an x^0 profile). $\phi(x^0, x^4)$ by the method of lines on n equally spaced grid points x_j^0 ($j = 0, \dots, n-1$, spacing h) in $[0.05, 0.22] \subset (0, \pi/12)$ ($H = 1$, i.e. $6Hx^0 \in [0.3, 1.32]$), in flux form with zero flux at both ends:

$$\begin{aligned} \ddot{\phi}_j &= \cos_j \frac{F_{j+1/2} - F_{j-1/2}}{h} - V'(\phi_j), & F_{j+1/2} &= c_{j+1/2} \frac{\phi_{j+1} - \phi_j}{h}, \\ c &= \sec \cot^2 \text{ at the half points,} & F_{-1/2} &= F_{n-1/2} = 0. \end{aligned} \quad (25)$$

Summation by parts shows that this semi-discrete scheme conserves exactly

$$E_h = \sum_j \sec_j h \left(\frac{1}{2} \dot{\phi}_j^2 + V(\phi_j) \right) + \sum_{j+1/2} \frac{1}{2} c_{j+1/2} \frac{(\phi_{j+1} - \phi_j)^2}{h}, \quad (26)$$

so its drift measures the time integrator alone. Sheet averages are weighted by the volume element $\sec 6Hx^0$: $w_{\text{sheet}} \equiv \langle P \rangle / \langle \rho \rangle$, never $\langle P/\rho \rangle$; the local w at the middle grid point is reported beside it. The initial profile is $\phi(x^0, 0) = \phi_i [1 + \varepsilon \cos(\pi(x^0 - x_{\min}^0)/(x_{\max}^0 - x_{\min}^0))]$ at rest.

Model E (fable on the pre-universe). $\partial_4 s = 0$ exactly (F3): nothing to integrate.

The CPL fit. For every run of Models A, A' and B: unweighted least squares of $w(a)$ to (1) over $0.3 \leq a \leq 1$ ($z \leq 2.33$), reported with the true min w , $w(a = 0.3)$, $w(1)$, the extrapolation $w_0 + w_a$, and the Euclidean distance of (w_0, w_a) from Unite's point. This fit is a description of a model curve, not a fit to supernova data.

6 Numerical method

The engine. The author's three rustSolveIt repositories [18, 19, 20] contain a rigid-body, planetary and quantum simulator built on a vendored pure-Rust translation of SUNDIALS 7.8.0 [14, 15] (`sundials_rs/crates/sundials_core` and `sundials_rs/crates/cvode_rs`). Their notebooks drive the simulator's own command language and cannot integrate a cosmological field equation, so they were not used as they stand; their engine was. The solver of this work, the Rust crate `fable-cosmology/rust/fable_cosmo`, depends on those two crates by path exactly as the repositories' own `planet_Mercury/mercury_rs` does, and pins `-C target-feature=+fma` in its `.cargo/config.toml`, which the engine's deterministic math library requires on x86-64. The engine is fetched by the setup script as a sparse clone (`sundials_rs` only) of the repository for the student's platform.

The integrator. CVODE with BDF, Newton iteration and a dense direct linear solver with a finite-difference Jacobian for the FLRW models; the Adams family for the undamped oscillations of Models C and D (`--method` overrides either); scalar tolerances $\text{rtol} = 10^{-10}$, $\text{atol} = 10^{-12}$ (rustSolveIt's defaults); `CVodeSetStopTime` at the end of the range; outputs at equally spaced abscissae in `CV_NORMAL` mode. Typical costs: Model A, `exp` $\lambda = 1$: 385 steps and 434 right-hand-side evaluations for 701 outputs; Model B: 811 steps and 869 evaluations; Model C `quadratic` and `quartic` to $x^4 = 100$: 965 and 2974 steps; Model D on 61 grid points to $x^4 = 30$: 294 095 steps ($\varepsilon = 0.05$) and 329 282 steps ($\varepsilon = 0.5$).

The solver's contract. One command-line program,

```
fable_cosmo <scalar-flrw|scalar-cpl|spinor-flrw|scalar-pre|scalar-pre-x0>
  [--potential NAME] [--param KEY=VALUE ...] [--n0 X0] [--n1 X1] [--points K]
  [--out FILE.csv] [--no-normalize] [--grid N] [--x0min A] [--x0max B]
  [--profile-out FILE.csv] [--rtol R] [--atol A] [--method bdf|adams]
```

writes one CSV row per output point with a header naming every column (the column lists are in Appendix A), prints the initial conditions, the CVODE statistics and the potential after normalisation on standard error, and exits with code 0 on success, 1 on a solver or physics error (with a one-line reason), 2 on a malformed command line.

Unit tests. Seven tests (`cargo test --release`): every potential’s derivative against a central finite difference (scalar and spinor); the spinor’s w formulas, including dust for the mass term, $w = n - 1$ for a power law, the crossings at s_1 and s_* , positivity of `lorentz` and `expdamp` and the invariance of w under $V \rightarrow kV$; the CPL closed form against the integral and $H_{\text{CPL}}^2(1) = 1$; $\rho + P = 2\text{KE}$; the Friedmann closure $H(N = 0) = 1$; the refusal of an x^0 grid that touches $x^0 = 0$ or $6Hx^0 = \pi/2$; and CVODE BDF and Adams against the exact exponential decay.

The notebooks. Four Jupyter notebooks, generated by a script and executed headlessly, drive the solver, write every result under `fable-cosmology/results/`, cross-check every run with `scipy`, and assert every number their text quotes: `01_fableScalar_quintessence` (Models A and A’), `02_fable_spinor` (Model B), `03_pre-universe_dynamics` (Models C and D, E stated) and `04_dark_matter_dark_energy` (the synthesis and the figures of Section 8.4). They follow the eight notebook requirements of the `rustSolveIt` repositories, adapted, and a checker verifies them.

7 Verification

Symbolic. Part VI of the Mathematica notebook (Sections 23–25, Input cells 153–172) adds 108 assertions to the 250 of Parts I–V. Evaluated headlessly from the notebook file, all 172 Input cells run with no cell raising a message and the tally reads *358 assertions, 358 passed, 0 failed*; no identity was accepted on numerical evidence alone, and each of the 27 non-vanishing claims is certified by a numerical witness. Part VI never gives a_4 a value, and its last assertion checks that a_4 is still undefined.

Three independent integrators. For one reference case of each field the same equations were integrated by CVODE (Rust), by `scipy`’s explicit Dormand–Prince DOP853 [16, 17] at $\text{rtol} = 10^{-11}$, $\text{atol} = 10^{-14}$, and by Mathematica’s `NDSolve` at 24-digit working precision (`reference/make_reference.wls`; Part VI repeats the `NDSolve` runs and agrees with the stored CSVs at every row to 10^{-9}). Table 3 gives the maximum differences over the 701 output points.

Table 3: Three integrators on the reference cases: maximum absolute differences over 701 points. Model A: `exp`, $\lambda = 1$, $V_0 = 2.6871520526047146$ (the solver’s normalised value). Model B: `expdamp` (V_0, m, s_1) = (1.566, 0.839, 2.21), not normalised.

quantity	Model A (scalar)	Model B (spinor)
$\max w_{\text{CVODE}} - w_{\text{NDSolve}} $	1.149×10^{-10}	1.861×10^{-8}
$\max w_{\text{scipy}} - w_{\text{NDSolve}} $	1.313×10^{-10}	1.008×10^{-11}
$\max w_{\text{CVODE}} - w_{\text{scipy}} $	9.509×10^{-11}	1.861×10^{-8}
$\max t_{\text{CVODE}} - t_{\text{NDSolve}} $	5.428×10^{-10}	5.224×10^{-9}
$w(a = 1)$, CVODE	−0.843363804027	−0.860845643423
$w(a = 1)$, <code>scipy</code>	−0.843363803932	−0.860845640379
$w(a = 1)$, <code>NDSolve</code>	−0.843363803931	−0.860845640378

For Model B the relative differences between CVODE and `NDSolve` are 5.2×10^{-8} in s , 2.5×10^{-9} in H and 1.4×10^{-8} in ρ_Ψ : the spinor’s difference in w is CVODE’s error on s at $\text{rtol} = 10^{-10}$ (a few parts in 10^8 after 7 e-folds of $s \propto e^{-3N}$) amplified by the slope dw/ds near the phantom dip.

Every run re-integrated. Each notebook re-integrates every one of its runs with scipy DOP853 from the same initial state and the same normalised potential and fails if any difference in w exceeds 10^{-6} . The largest differences are 4.1×10^{-9} over the 13 runs of Models A and A', 2.4×10^{-8} over the 11 runs of Model B, 2.2×10^{-9} for Model C **quadratic**, 6.5×10^{-9} and 1.07×10^{-8} in w_{sheet} for the two Model D runs (a 122-equation system re-written in numpy), and 4.0×10^{-7} for Model C **quartic**. The last was investigated rather than accepted: repeating the CVODE run at $\text{rtol} = 10^{-12}$, $\text{atol} = 10^{-14}$ reduces it to 2.5×10^{-9} (and to 8.6×10^{-10} against scipy at $\text{rtol} = 10^{-13}$), which identifies it as CVODE's accumulated phase error over the ~ 13 nonlinear periods at the default tolerance.

Physics sanity. The **const** potential gives $w = -1$ at every point ($\max|w + 1| = 0$) with $H(a = 1) = 1$ to 1.1×10^{-14} ; the **mass** spinor gives $w = 0$ at all 701 points (asserted to 10^{-14} ; the stored values are exactly 0) and $q_{\text{dec}}(a = 1) = 0.500042$; the numerical crossings of $w = -1$ agree with $a_c = (s_1/s_0)^{-1/3}$ to 1.5×10^{-8} (Table 5); Model C conserves ρ to 3.3×10^{-9} (**quadratic**) and 1.6×10^{-8} (**quartic**) over 100 time units and reproduces $\phi = \cos x^4$ to 1.7×10^{-9} ; Model D conserves E_h to 4.5×10^{-9} and 9.1×10^{-9} .

8 Results

8.1 fableScalar in the reference model (Models A and A')

The exponential potential. With the frozen start, the normalised runs $\lambda = 0.5, 1, 1.5, 2$ reach $w(1) = -0.9616, -0.8434, -0.6322, -0.2720$ with $\Omega_\phi(1) = 0.699916$ and $\min w = -1.000000$ in every case (Figure 1); all four thaw. For $\lambda = 3$ ($\lambda^2 > 3$) the late-time attractor is the scaling solution, on which the field tracks the dominant component with $\Omega_\phi = 3/\lambda^2$ [11]; the normalisation cannot bracket its target ($\Omega_\phi(10^6 V) = 0.3375 < 0.6999$) and the solver refuses. Run by hand with $V_0 = 10^6$ it gives $\Omega_\phi(1) = 0.3375$ ($3/\lambda^2 = 0.3333$), $w(1) = -0.0039$ and $q_{\text{dec}}(1) = +0.4981$: not a dark-energy model.

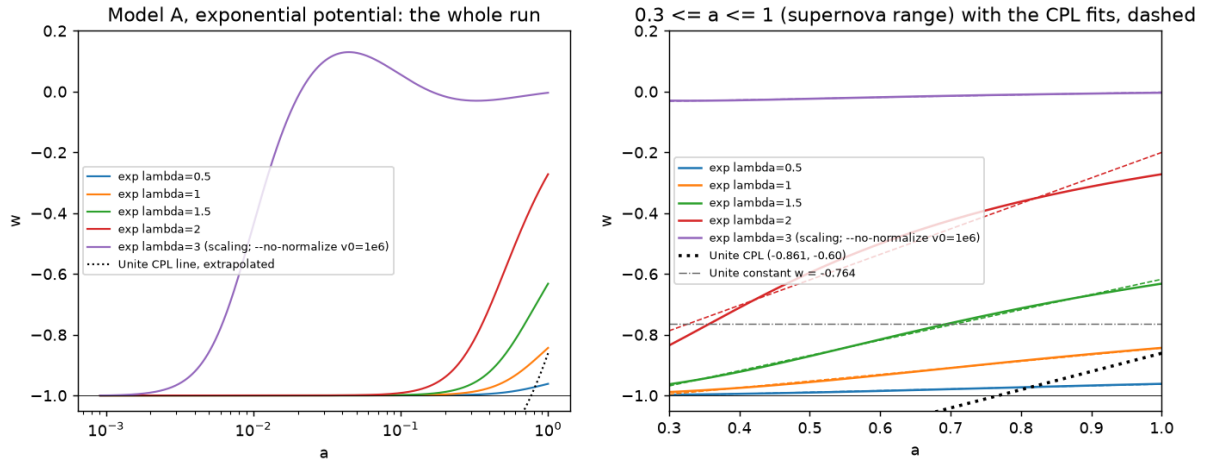


Figure 1: Model A, exponential potential. Left: $w(a)$ over the whole run, with Unite's CPL line extrapolated (dotted). Right: the supernova range $0.3 \leq a \leq 1$ with each run's CPL fit (dashed), Unite's line (dotted) and the constant- w fit -0.764 (dash-dotted). The $\lambda = 3$ run is the scaling solution with $V_0 = 10^6$ set by hand.

The other potentials. The inverse power ($\alpha = 1$) does not join its tracker ($w = -2/3$ in the matter era) from a frozen start: starting at $N_0 = -7$ or at $N_0 = -25$ gives the same history to 3.0×10^{-8} in w , because Hubble friction in the radiation era holds the field frozen

($w(N = -4) = -0.99975$, $w(N = -2) = -0.92110$); it thaws late, overshoots and turns down, $w(1) = -0.7532$, with fit $(-0.7365, +0.0625)$, class *mixed*. The PNGB potentials thaw mildly ($f = 1$: $(-0.9888, -0.0165)$; $f = 0.5$ from $\phi_i = 0.3$: $(-0.8841, -0.1817)$), the hilltop with $\mu = 3$ is almost frozen ($(-0.9993, -0.0010)$), and the control is $w = -1$ to machine precision (Figure 2).

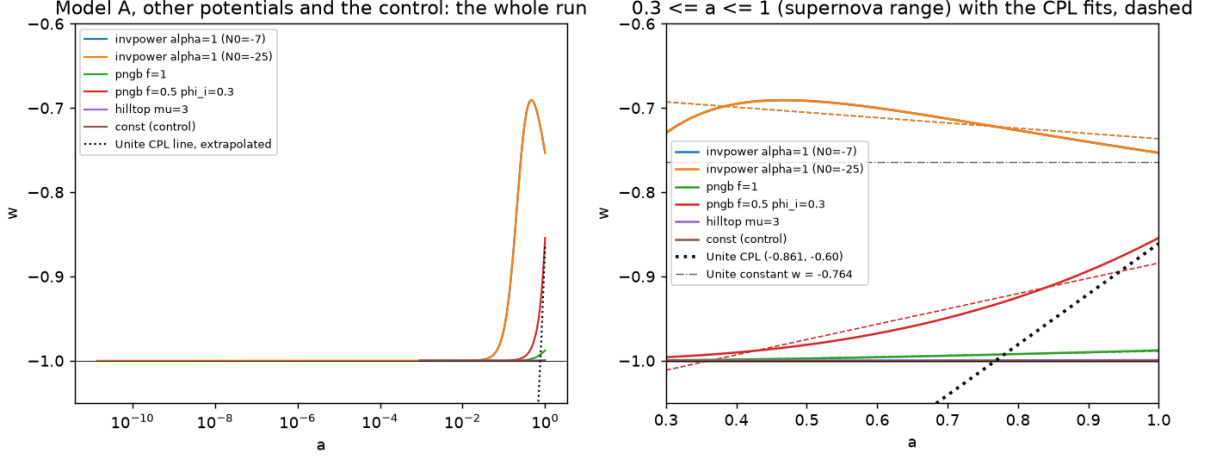


Figure 2: Model A, the inverse-power (two starts, indistinguishable), PNGB, hilltop and constant potentials; panels as in Figure 1.

Model A'. As a test field on the Unite CPL background, $\lambda = 1$ gives $w(1) = -0.8391$ and fit $(-0.8371, -0.2257)$, essentially the self-consistent result. $\lambda = 2$ cannot carry $\Omega_{DE} = 0.70$ ($\rho_\phi(10^6 V) = 1.025 < 2.100$): a test field with $\lambda^2 > 3$ scales with the background; run by hand with $V_0 = 10$ it carries $\Omega = 0.4984$ today, with $w(1) = -0.2980$ (Figure 3).

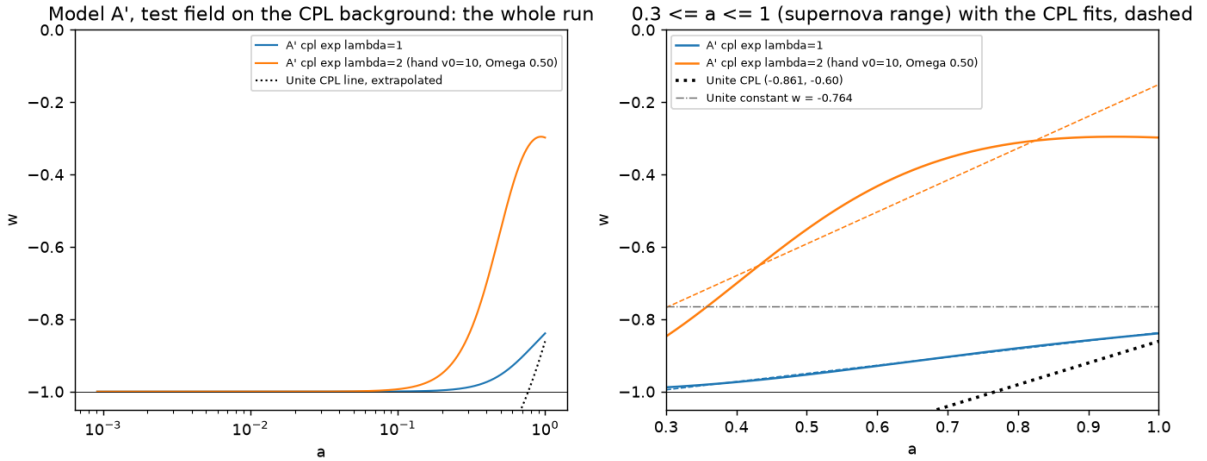
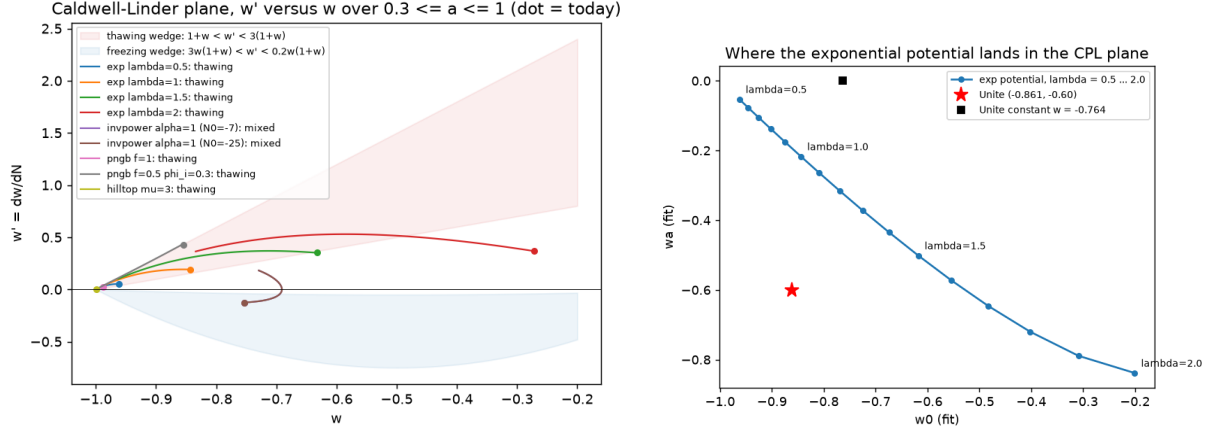


Figure 3: Model A': the exponential fableScalar as a test field on the Unite CPL background ($\lambda = 1$ normalised; $\lambda = 2$ with $V_0 = 10$ by hand, $\Omega = 0.50$ today).

Thawing, and where the exponential lands in the CPL plane. In the Caldwell–Linder plane (Figure 4, left) every exponential, PNGB and hilltop run stays above $w' = dw/dN = 0$ over the fit range: thawing; the inverse-power runs cross the axis: mixed. A scan $\lambda = 0.5, 0.6, \dots, 2.0$ (Figure 4, right; `results/nb01_lambda_scan.csv`) moves the fit monotonically from $(-0.9627, -0.0532)$ to $(-0.2004, -0.8384)$. The distance from Unite's point is smallest at $\lambda = 1.4$, $(w_0, w_a) = (-0.6740, -0.4349)$, distance 0.2495; the fitted w_a reaches -0.60 at $\lambda = 1.638$, where $w_0 = -0.5274$. Table 4 lists the fits.



(a) The Caldwell–Linder plane w' versus w , $0.3 \leq a \leq 1$; (b) The fitted (w_0, w_a) of the exponential potential for $\lambda = 0.5 \dots 2.0$. dots mark today.

Figure 4: Model A: classification and the path of the CPL fit.

Table 4: CPL fits over $0.3 \leq a \leq 1$ of Models A and A' (results/nb01_cpl_fits.csv). $\min w = -1$ for every run; “dist” is the distance of (w_0, w_a) from Unite’s point.

run	w_0	w_a	$w_0 + w_a$	$w(0.3)$	$w(1)$	class	dist
exp $\lambda = 0.5$	-0.9627	-0.0532	-1.0159	-0.9976	-0.9616	thawing	0.556
exp $\lambda = 1$	-0.8440	-0.2172	-1.0612	-0.9887	-0.8434	thawing	0.383
exp $\lambda = 1.5$	-0.6176	-0.5015	-1.1191	-0.9632	-0.6322	thawing	0.263
exp $\lambda = 2$	-0.2004	-0.8384	-1.0388	-0.8355	-0.2720	thawing	0.702
exp $\lambda = 3$ (scaling, $\Omega = 0.34$)	-0.0020	-0.0427	-0.0448	-0.0299	-0.0039	mixed	1.024
invpower $\alpha = 1$ ($N_0 = -7$)	-0.7365	+0.0625	-0.6741	-0.7298	-0.7532	mixed	0.674
pngb $f = 1$	-0.9888	-0.0165	-1.0053	-0.9994	-0.9878	thawing	0.597
pngb $f = 0.5$, $\phi_i = 0.3$	-0.8841	-0.1817	-1.0657	-0.9959	-0.8542	thawing	0.419
hilltop $\mu = 3$	-0.9993	-0.0010	-1.0003	-1.0000	-0.9992	thawing	0.615
const (control)	-1.0000	0.0000	-1.0000	-1.0000	-1.0000	constant	0.616
A' exp $\lambda = 1$	-0.8371	-0.2257	-1.0628	-0.9883	-0.8391	thawing	0.375
A' exp $\lambda = 2$ ($\Omega = 0.50$)	-0.1515	-0.8806	-1.0321	-0.8476	-0.2980	mixed	0.763
Unite CPL	-0.861	-0.60	-1.461	-1.281	-0.861		0
Unite constant w	-0.764	0	-0.764	-0.764	-0.764		0.608

8.2 fable in the reference model (Model B)

Dust and Λ . The mass spinor has $w = 0$ at every point and $\rho_\Psi a^3$ constant to 5.3×10^{-8} (CVODE's error on s). `lambda-mass` with $V_0 = 0.7$, $m = 0.3$ before normalisation (1.469824 and 0.629924 after) splits into $\Omega_\Lambda = V(0)/3 = 0.489941$, a bare constant in the action, and spinor dust $\Omega_{\text{dust}} = ms_0/3 = 0.209975$; $w(1) = -V_0/(V_0 + ms_0) = -0.700000$. It is Λ CDM with the dust supplied by the spinor. The `power` potentials with $m = \lambda$ are dust early and tend to $n - 1$ late: $w(1) = -0.38200, -0.25000, -0.05000$ for $n = 0.236, 0.5, 0.9$ ($w(a = 0.1) = -0.00388, -0.01533, -0.03339$); run into the future to $N = +6$ the $n = 0.236$ case reaches $w = -0.7639919$, Unite's constant value to 8.1×10^{-7} (Figures 5, 6).

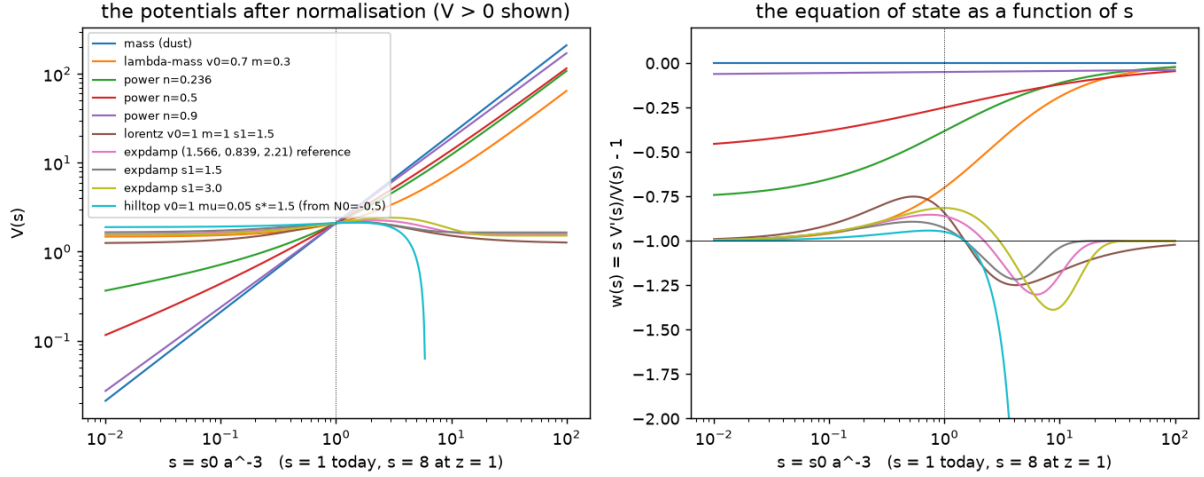


Figure 5: The potentials of fable after normalisation (left, $V > 0$ shown) and $w(s) = sV'/V - 1$ (right), against $s = s_0 a^{-3}$; $s = 1$ today, $s = 8$ at $z = 1$. Where V' changes sign w crosses -1 .

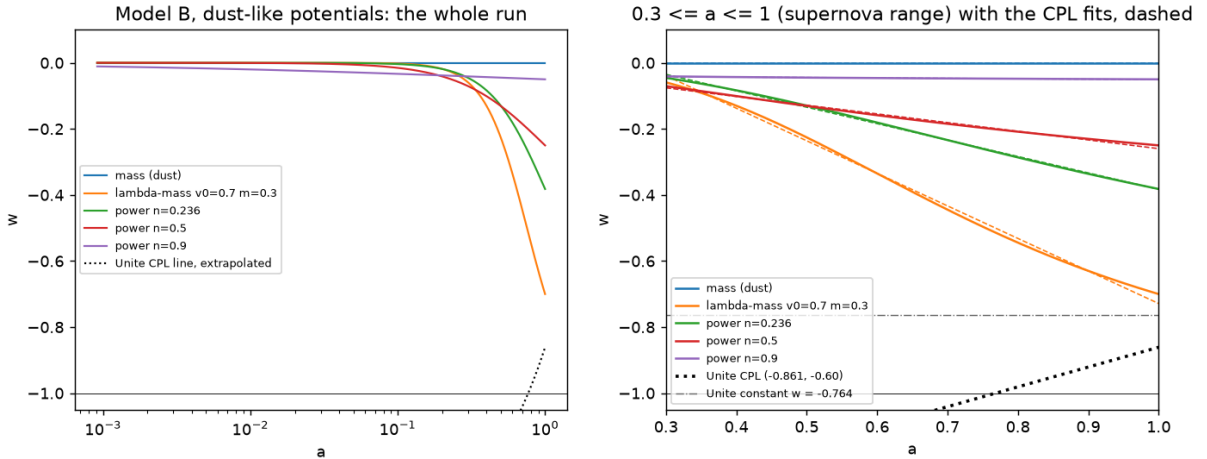


Figure 6: Model B, the dust-like potentials: mass, lambda-mass and power; panels as in Figure 1.

Crossing the phantom divide. `lorentz`, `expdamp` (three values of s_1) and `hilltop` all cross $w = -1$ exactly once, at the analytic scale factor (Table 5, Figure 7). The reference `expdamp` (1.566, 0.839, 2.21) has $w(1) = -0.86085$, crosses -1 at $a_c = 0.7677$ ($z = 0.30$), reaches $w = -1.30210$ at $a = 0.5434$ ($z = 0.840$) and tends to -1 from below as $a \rightarrow 0$; V never falls below $V_0 = 1.566$, so $\rho_\Psi > 0$ throughout. The `hilltop` is refused from the default start ($V(s) = -1.8 \times 10^{17}$ at $s = 1.3188 \times 10^9$) and is run only inside its window $s < s_* + \sqrt{V_0/\mu} = 5.9721$ ($a > 0.5512$), from $N_0 = -0.5$.

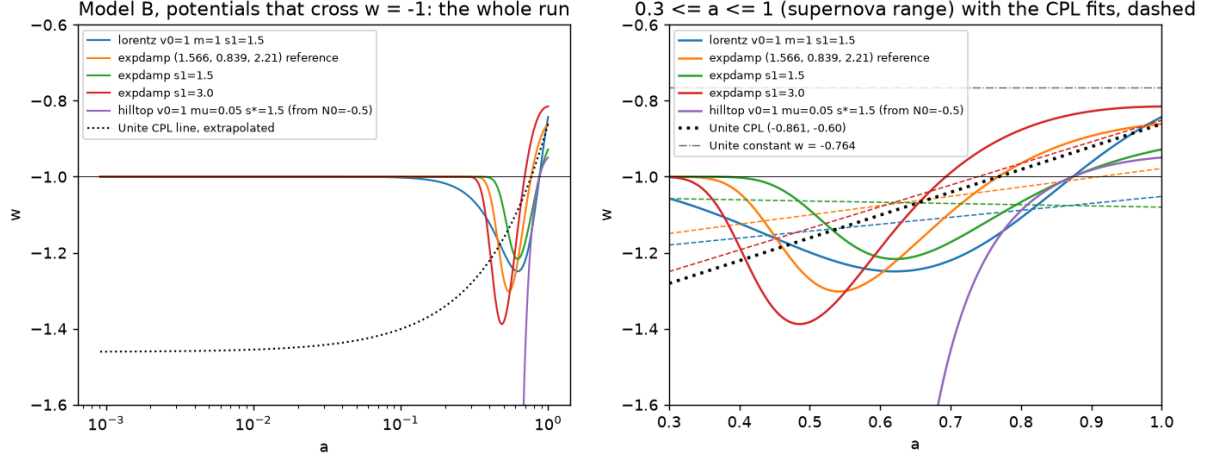


Figure 7: Model B, the potentials that cross $w = -1$: lorentz, expdamp (reference, $s_1 = 1.5$, $s_1 = 3.0$) and hilltop (from $N_0 = -0.5$); panels as in Figure 1.

Table 5: Crossings of $w = -1$ in Model B, numerical (local degree-6 polynomial and root bracket on the 701-point grid) against the analytic $a_c = (s_1/s_0)^{-1/3}$ (`results/nb02_crossings.csv`).

run	a_c numerical	a_c analytic	$ \Delta $	min V
lorentz $V_0 = m = 1$, $s_1 = 1.5$	0.8735804797	0.8735804647	1.50×10^{-8}	1.2408
expdamp (1.566, 0.839, 2.21) reference	0.7677195193	0.7677195065	1.29×10^{-8}	1.5660
expdamp $s_1 = 1.5$	0.8735804797	0.8735804647	1.50×10^{-8}	1.6468
expdamp $s_1 = 3.0$	0.6933612858	0.6933612744	1.15×10^{-8}	1.5173
hilltop $s_* = 1.5$ (from $N_0 = -0.5$)	0.8735804648	0.8735804647	4.00×10^{-11}	1.1811

Table 6: CPL fits over $0.3 \leq a \leq 1$ of Model B (`results/nb02_cpl_fits.csv`). The hilltop run starts at $a = 0.61$, so its fit covers $0.61 \leq a \leq 1$ and its $w(0.3)$ is undefined.

run	w_0	w_a	$w_0 + w_a$	$w(0.3)$	$w(1)$	min w	class	dist
mass	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	constant	1.049
lambda-mass	-0.7286	+0.9850	+0.2564	-0.0593	-0.7000	-0.7000	freezing	1.591
power $n = 0.236$	-0.3830	+0.4971	+0.1141	-0.0455	-0.3820	-0.3820	freezing	1.197
power $n = 0.5$	-0.2598	+0.2630	+0.0032	-0.0706	-0.2500	-0.2500	freezing	1.052
power $n = 0.9$	-0.0508	+0.0128	-0.0381	-0.0411	-0.0500	-0.0500	freezing	1.016
lorentz	-1.0519	-0.1826	-1.2346	-1.0570	-0.8427	-1.2492	mixed	0.459
expdamp reference	-0.9782	-0.2445	-1.2227	-1.0000	-0.8608	-1.3021	mixed	0.374
expdamp $s_1 = 1.5$	-1.0801	+0.0322	-1.0479	-1.0000	-0.9281	-1.2166	mixed	0.669
expdamp $s_1 = 3.0$	-0.8513	-0.5692	-1.4205	-1.0010	-0.8151	-1.3878	mixed	0.032
hilltop (from $a = 0.61$)	-0.5265	-4.0285	-4.5550	—	-0.9494	-3.4057	thawing	3.445

The straight line is a poor description of a $w(a)$ that dips to -1.3 and comes back (Table 6): the reference `expdamp` fits $(-0.9782, -0.2445)$ although its $w(1)$ is -0.861 . In the Caldwell–Linder plane (Figure 8) the phantom runs sit at $w < -1$ in the past and rise through the divide towards today, so their dw/dN changes sign over the fit range: mixed. The dust-like runs are freezing.

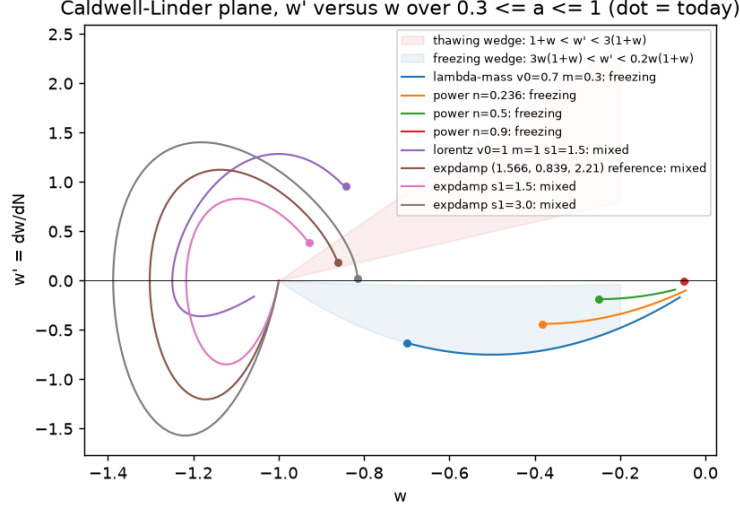


Figure 8: Model B in the Caldwell–Linder plane, $0.3 \leq a \leq 1$; dots mark today.

8.3 fableScalar on the pre-universe (Models C and D)

Model C. The quadratic potential ($m = 1$, $\phi_i = 1$, to $x^4 = 100$, 2001 outputs) reproduces the exact solution $\phi = \cos x^4$ to 1.67×10^{-9} and $w = -\cos 2x^4$ to 2.05×10^{-9} ; w swings between -1 and $+1$ while $\rho = \frac{1}{2}$ is conserved to 3.27×10^{-9} , and the running average is $w_{\text{avg}}(100) = +0.004363$ (Figure 9). The quartic potential ($\lambda = 1$) averages to $w_{\text{avg}}(100) = 0.334909$, within 1.6×10^{-3} of the virial value $1/3$: the oscillating field averages to radiation; ρ is conserved to 1.58×10^{-8} (Figure 10). The residuals of the averages are the last incomplete period divided by the length of the run.

Model D: the sloshing mechanism. The quadratic potential on 61 grid points in $x^0 \in [0.05, 0.22]$, from a profile at rest, to $x^4 = 30$ (Figures 11, 12). Because the flux coefficient $\sec \cot^2$ is large near $x^0 = 0.05$ ($\cot^2 0.3 = 10.4$), even a 5 % ripple ($\varepsilon = 0.05$) stores 28.5 % of E_h in G_0 at the start. The gradient store exchanges energy with the x^4 motion: for $\varepsilon = 0.05$ the G_0 fraction ranges over $[0.0288, 0.2847]$ and w_{sheet} over $[-1, +0.9412]$; for $\varepsilon = 0.5$, 97.8 % of E_h starts in G_0 , the fraction falls to 9.88 % and comes back, and w_{sheet} ranges over $[-1, +0.8022]$. Because the quadratic problem is linear, the virial theorem holds for the whole sheet, $\langle \text{KE} \rangle = \langle \text{PE} \rangle + \langle G_0 \rangle$: the time-averaged kinetic fractions are 0.5017 and 0.4997 and the running averages of w_{sheet} at $x^4 = 30$ are $+0.0038$ and $+0.0007$. The gradient store changes the history of w , not its average; only a massless profile stays at $w = -1$. E_h is conserved to 4.48×10^{-9} and 9.05×10^{-9} .

Model E. For fable on the pre-universe nothing needs integrating: $\partial_4 s = 0$ exactly (F3), so V , ρ_Ψ and w_Ψ are constant along the observer’s time. The a^{-3} dilution of Model B is the reference model’s Hubble term, which the pre-universe does not have.

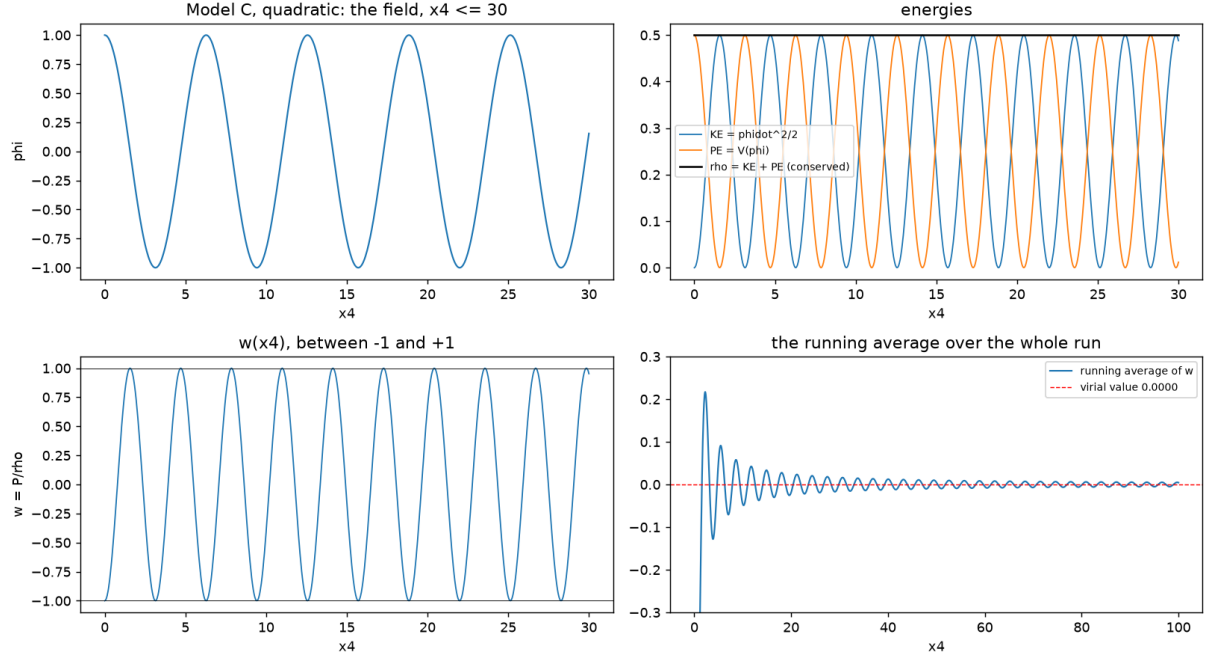


Figure 9: Model C, quadratic potential, on the pre-universe: no Hubble friction. The field, the exchange of kinetic and potential energy at constant ρ , $w(x^4)$ between -1 and $+1$, and its running average settling on the virial value 0.

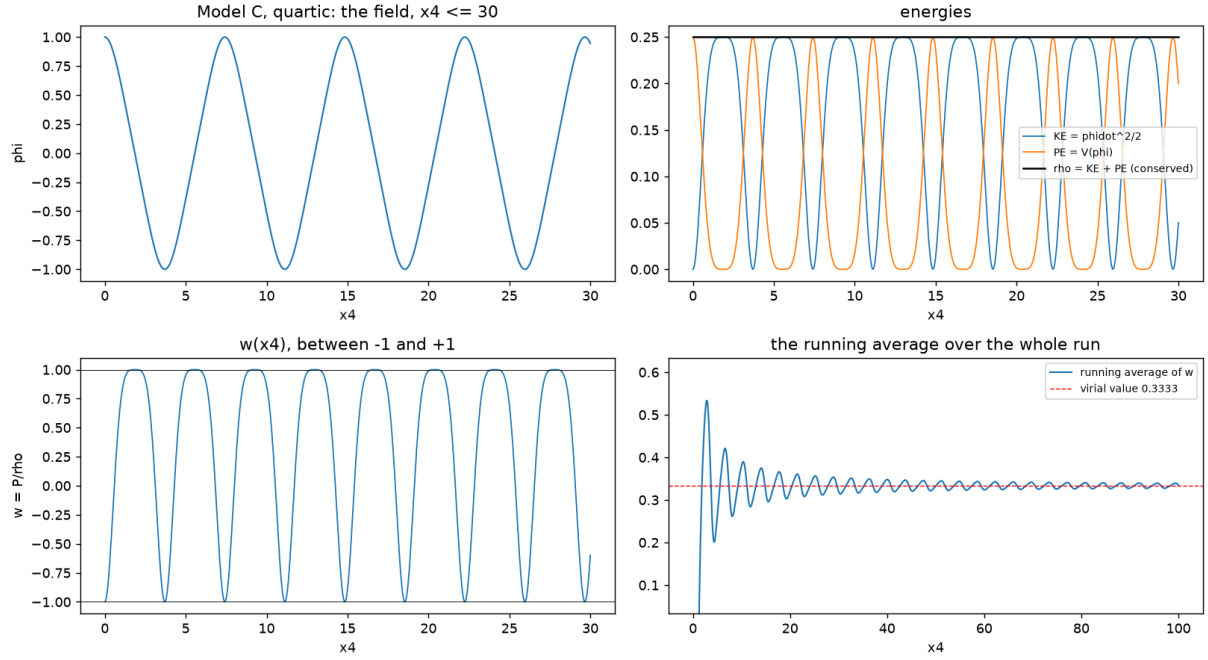


Figure 10: Model C, quartic potential: the running average of w settles on the virial value $1/3$.

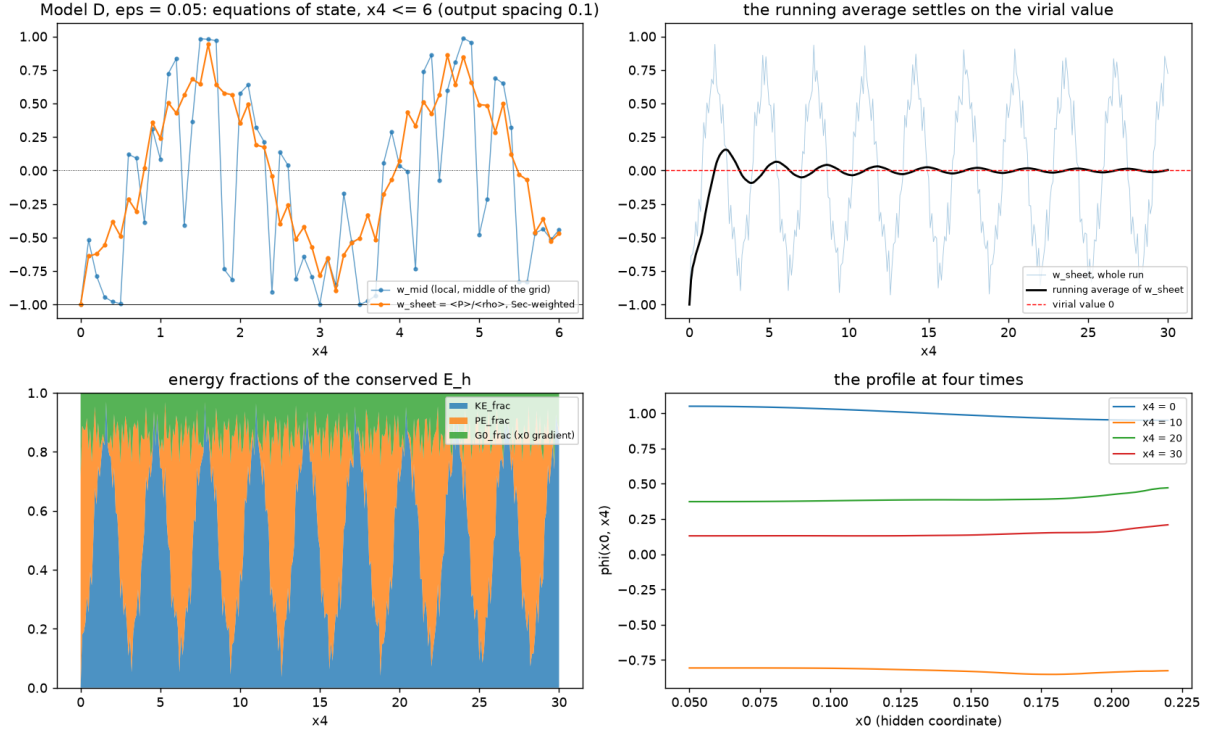


Figure 11: Model D, $\varepsilon = 0.05$: the local w_{mid} and the sec-weighted $w_{\text{sheet}} = \langle P \rangle / \langle \rho \rangle$ over $x^4 \leq 6$ (dots are the outputs, spacing 0.1); the running average of w_{sheet} ; the energy fractions of the conserved E_h ; the profile $\phi(x^0)$ at four times.

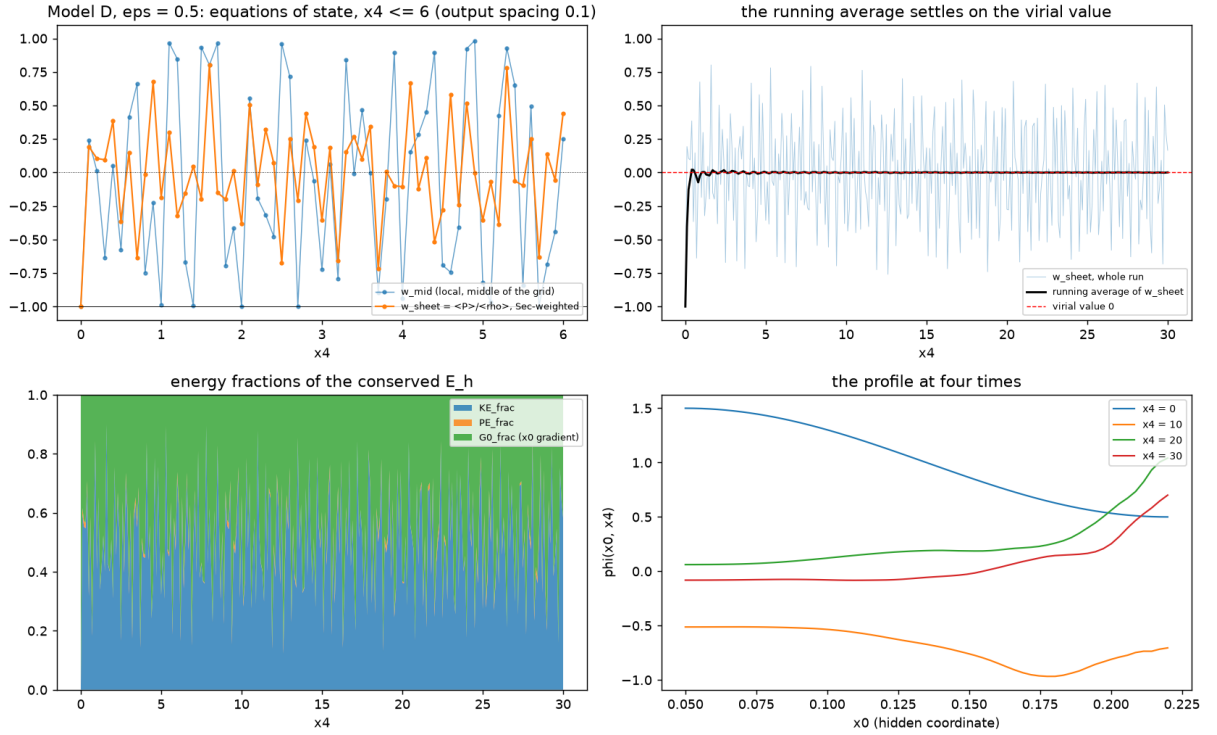


Figure 12: Model D, $\varepsilon = 0.5$; panels as in Figure 11. Most of the energy starts in the x^0 gradient, a $w = -1$ store on the observed sheet, and sloshes into the x^4 motion and back.

8.4 The synthesis

Figure 13 shows the best cases of both fields against Unite's line. Figure 14 shows the density fractions: at the start radiation is $\Omega_r = 0.2349$; the field overtakes matter at $a = 0.7376$ ($z = 0.356$) for the reference **expdamp** spinor and at $a = 0.7239$ ($z = 0.381$) for the $\lambda = 1$ scalar. Figure 15 shows the deceleration parameter: acceleration begins ($q_{\text{dec}} = 0$) at $a = 0.5475$ ($z = 0.827$) for the reference **expdamp**, $a = 0.5840$ ($z = 0.712$) for the $\lambda = 1$ scalar and $a = 0.8045$ ($z = 0.243$) for **lambda-mass**, and never for the **mass** spinor ($q_{\text{dec}}(1) = +0.5000$). Today $q_{\text{dec}} = -0.4037, -0.3854, -0.2349$ for these three, -0.3557 for **expdamp** $s_1 = 3.0$, -0.1637 for **exp** $\lambda = 1.5$ and $+0.0990$ for **power** $n = 0.236$ with $m = \lambda$, which does not accelerate today. The age today is $t(1) = 0.9472/H_0$ for the $\lambda = 1$ scalar, $0.9643/H_0$ for the reference **expdamp**, $0.9637/H_0$ for the **const** control and $0.6666/H_0$ for the **mass** spinor (all dust: the Einstein–de Sitter $2/3$).

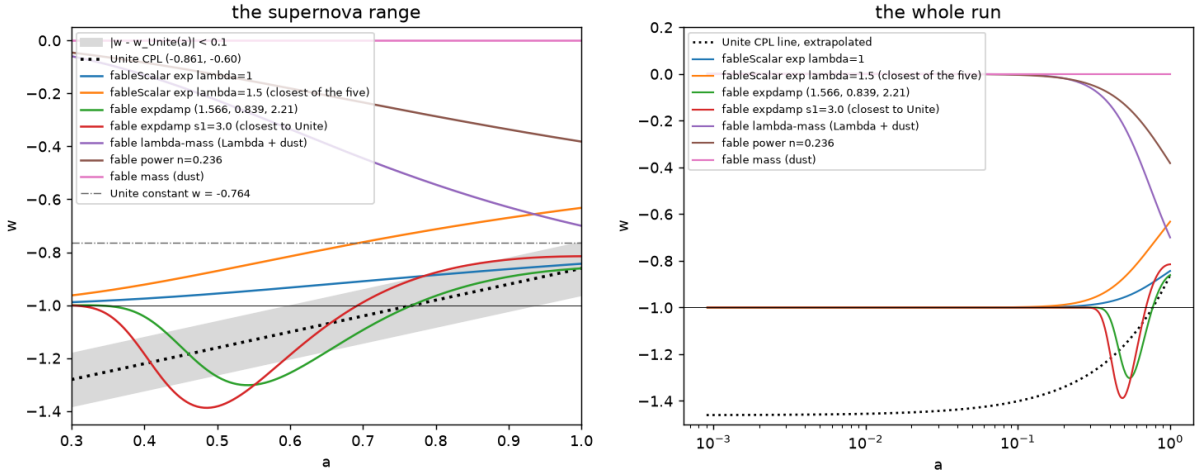


Figure 13: The best cases of both fields. Left: the supernova range with Unite's CPL line and the band $|w - w_{\text{Unite}}(a)| < 0.1$. Right: the whole run with the CPL line extrapolated.

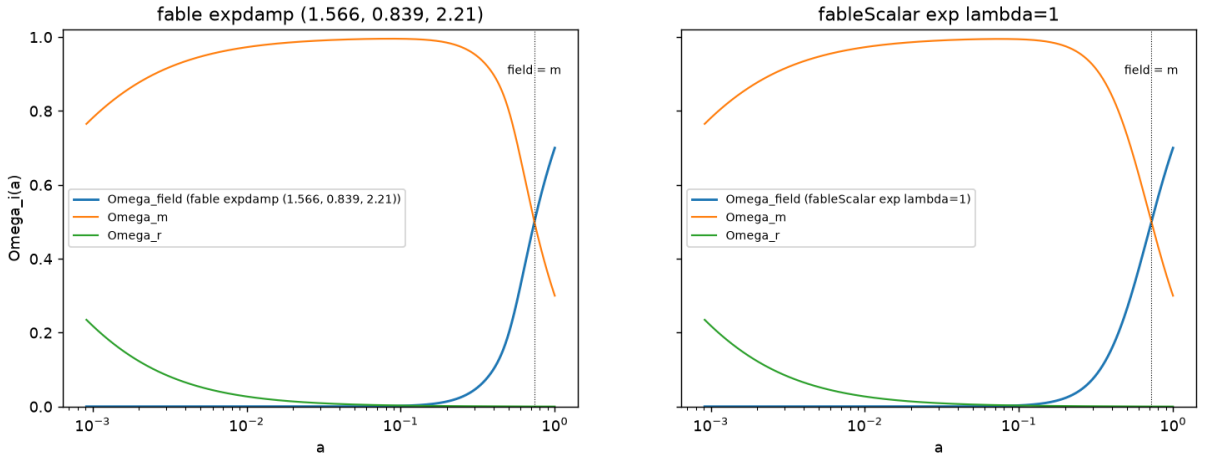


Figure 14: Density fractions Ω_{field} , Ω_m , Ω_r for the reference **expdamp** spinor (left) and the $\lambda = 1$ exponential scalar (right); the dotted line marks where the field overtakes matter.

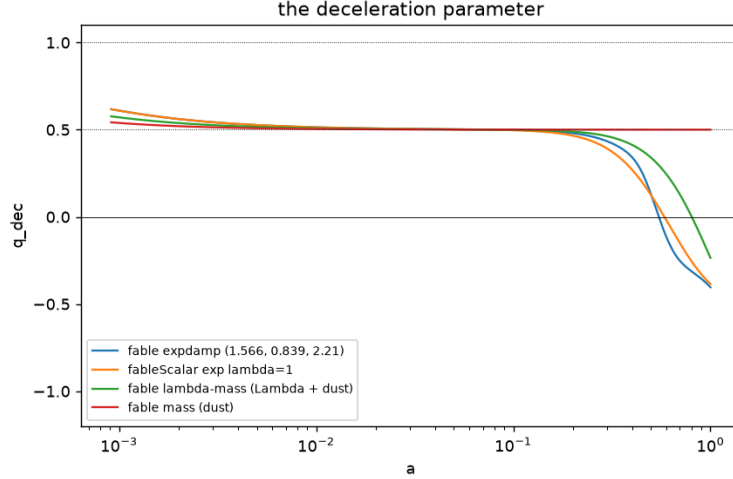


Figure 15: The deceleration parameter $q_{\text{dec}} = -1 - (dH/dN)/H$.

9 Comparison with the Supernovae Unite fits

Table 7: The runs closest to Unite’s $(w_0, w_a) = (-0.861, -0.60)$, sorted by distance, with the constant- w reference. All fits over $0.3 \leq a \leq 1$.

run	model	w_0	w_a	$w_0 + w_a$	$w(1)$	min w	dist
Unite CPL		-0.861	-0.60	-1.461	-0.861		0
expdamp $s_1 = 3.0$	B	-0.8513	-0.5692	-1.4205	-0.8151	-1.3878	0.032
exp $\lambda = 1.4$ (scan)	A	-0.6740	-0.4349	-1.1089	-0.6832	-1.0000	0.250
exp $\lambda = 1.5$	A	-0.6176	-0.5015	-1.1191	-0.6322	-1.0000	0.263
expdamp reference	B	-0.9782	-0.2445	-1.2227	-0.8608	-1.3021	0.374
exp $\lambda = 1$ on the CPL background	A'	-0.8371	-0.2257	-1.0628	-0.8391	-1.0000	0.375
exp $\lambda = 1$	A	-0.8440	-0.2172	-1.0612	-0.8434	-1.0000	0.383
lorentz	B	-1.0519	-0.1826	-1.2346	-0.8427	-1.2492	0.459
Unite constant w		-0.764	0	-0.764	-0.764		0.608
power $n = 0.236$, late times	B	$w \rightarrow n - 1 = -0.764$; $w(N = 6) = -0.76399919$					

Table 7 collects the comparison. Three conclusions follow.

A *thawing fableScalar* gives Unite’s sign pattern but not the pair. Every exponential run of the scan $\lambda = 0.5, \dots, 2.0$ has $w_0 > -1$ and $w_a < 0$, and its CPL extrapolation $w_0 + w_a$ lies below -1 (between -1.016 and -1.129 ; -1.06 for $\lambda = 1$, -1.12 for $\lambda = 1.5$), while the true w never drops below -1 (min $w = -1.000000$, the null energy condition). The “phantom past” of such a fit is the straight line leaving a curved $w(a)$ that is concave over the supernova range (it rises slowly from -1 and then faster), not a phase with $w < -1$. $|w_a| = 0.6$ is reachable ($\lambda = 1.638$) but then $w_0 = -0.53$; $w_0 = -0.861$ is reachable ($\lambda = 1$: -0.844) but then $w_a = -0.22$. The best compromise, $\lambda = 1.4$, is at distance 0.25.

A *fable spinor* with a sign-changing slope reproduces both numbers. The **expdamp** potential with $s_1 = 3.0$ fits $(-0.8513, -0.5692)$, $w_0 + w_a = -1.42$, at distance 0.032 from $(-0.861, -0.60, -1.46)$, with a genuine phantom phase (min $w = -1.388$, crossing at $a_c = 0.6934$, $z = 0.44$) and $\rho_\Psi = V \geq V_0 > 0$ throughout. Its true $w(1) = -0.815$ differs from the fitted w_0 because $w(a)$ is strongly curved. The reference parameters $(1.566, 0.839, 2.21)$ instead reproduce $w(1) = -0.8608$, Unite’s w_0 to three decimals, with a fit $(-0.978, -0.245)$.

Unite’s constant $w = -0.764$ is reproduced exactly, but only asymptotically, by the **power** spinor with $n = 0.236$; with $m = \lambda$ it is still $w(1) = -0.382$ today.

10 Dark matter and dark energy

Is there a connection to dark matter? Yes, and one of them is exact. (i) The spinor’s mass term, the author’s own $V = -(2M/H)s$ promoted to a cosmological field, is dust: $w_\Psi = 0$ identically (F1), and the **mass** run gives $w = 0$ at all 701 points with $\rho_\Psi \propto a^{-3}$ in the reference model. A 16-component spinor of $\text{Spin}(4,4)$ with a mass term is cold dark matter. (ii) A scalar oscillating in a quadratic potential on the pre-universe has the virial average $\langle w \rangle = 0$ (measured +0.0044), and in the reference model an oscillating field dilutes as dust. (iii) The **power** spinor with $n = 0.236$ is dust early ($w(a = 0.1) = -0.004$) and turns into dark energy later.

Is there a connection to dark energy? Yes, by two different mechanisms. **fableScalar** rolling under Hubble friction (Model A) is standard thawing quintessence, with Unite’s sign pattern and no phantom phase (Section 9). **fable** with a potential whose slope changes sign (Model B) gives a genuine phantom phase with positive energy and no wrong-sign kinetic term, the spinor-quintom mechanism [10], and comes within 0.032 of Unite’s fitted point. Two further spinor cases are dark energy of a different kind: **lambda-mass** is ΛCDM with spinor dust ($\Omega_\Lambda = 0.490$ is a bare constant in the action), and **power** $n = 0.236$ is field-driven, with no constant in the action, but reaches -0.764 only asymptotically.

What is the framework’s own mechanism? On the pre-universe itself there is no Hubble friction along the observer’s time, because the observed sheet’s expansion is compensated by the second sheet’s contraction. Hence **fableScalar** oscillates undamped with a conserved ρ and a virial-averaged w (0 for the mass term, $1/3$ for the quartic), and **fable**’s bilinear does not dilute at all, so its $w = sV'/V - 1$ is fixed once and for all. A field with a profile along the hidden coordinate stores energy in the x^0 gradient, which on the observed sheet is a $w = -1$ component and along x^0 an anisotropic stress; with a potential the store sloshes into the x^4 motion and back (Model D: up to 97.8% of the energy in the gradient, w_{sheet} between -1 and $+0.80$). A time-varying w on the pre-universe therefore comes from oscillation and sloshing, not from rolling. The average for a massive field is dust, and only a massless profile stays at $w = -1$.

11 Limitations

The following are stated so that nothing in this paper is read as more than it is. Most were found by an adversarial review of the design before any code was written, and they are corrections of that design.

1. **No gravitational sector.** The framework has no Einstein equations and a_4 is undetermined, so nothing here says which density gravitates or what $a(t)$ is. Every $H(a)$, every Ω and every CPL fit belongs to the 4-dimensional reference models, obtained by holding the hidden volume fixed and letting the field source a Friedmann equation on a separate FLRW frame. Their $a(t)$ is never substituted for a_4 .
2. **The kinematic identification** $\ln a = -a_4(Ht)$ is kinematics only; expansion of the observed sheet ($a'_4 < 0$) is a hypothesis about a sign, not a result.
3. **The volume bookkeeping.** The reduced density $\rho_4 \propto a^{-3}\rho_8$ dilutes like dust for every w . This is a volume effect, not a dark-matter signature (a cosmological constant passes the same test). The reduced tensor is not separately conserved, and the locally measured $T_{44} = \rho_8$ is exactly constant.
4. **Perturbations are not examined.** The spinor’s crossing of the phantom divide is a statement about the homogeneous background. The stability of perturbations in that phase, the known weak point of quintom models, has not been studied.

5. $w_\phi \geq -1$ **needs** $\rho_\phi > 0$. The sector with gradients along the timelike second sheet has negative energy and is excluded from every statement about w .
6. **The static x^0 profile** is Λ -like only on the observed sheet; along x^0 it is an anisotropic stress ($P_0 = +\rho$).
7. **The spinor’s homogeneous solutions.** A spinor homogeneous in every coordinate but x^4 is not a solution; the “no dilution” statement is for the rescaled field $\Psi' = \Psi/\sqrt{\sin 6Hx^0}$; $\Psi'(x^4)$ is exact only for constant V' ; for a nonlinear V the exact problem is the (x^0, x^4) system, and $\rho_\Psi = V + K_h$ carries a hidden-direction kinetic bilinear that vanishes only when Ψ' is x^0 -independent.
8. **Frame-dependent statements.** That the spin connection drops out of the Lagrangian is a property of the canonical frame (no totally antisymmetric connection); the product form of the divergence term rests on the anticommutator $\{\gamma^\mu, \Gamma_\mu\} = 0$ there. The energy-momentum tensor must be built with D_ν , not ∂_ν .
9. **Potentials.** The spinor hilltop is unbounded below and valid only inside its window; it cannot produce the Unite-like trajectory while $\rho > 0$, which is why **lorentz** and **expdamp**, both bounded below, are the phantom-crossing examples. The solver refuses every run with $V(s) \leq 0$. **lambda-mass** is Λ CDM with spinor dust, not field-driven dark energy. The **power** ordering (dust early, $n - 1$ late) needs $n < 1$.
10. **Initial conditions and classes.** The scalar starts frozen at a stated ϕ_i ; the inverse-power field does not reach its tracker from that start. Thawing and freezing are read off dw/dN , not assumed. Exponential potentials with $\lambda^2 > 3$ cannot carry $\Omega = 0.70$ today and are shown only as scaling solutions with a hand-set scale.
11. **The CPL fits** are unweighted least squares of model curves over $0.3 \leq a \leq 1$, not fits to supernova data; “distance” is Euclidean in the (w_0, w_a) plane and has no statistical meaning.
12. **Numerics.** The age offset $t_{\text{age}}(N_0) = 1/(2H(N_0))$ assumes radiation domination; since matter already contributes at $N_0 = -7$ the true age there lies between this and $2/(3H(N_0))$, an uncertainty of about $7 \times 10^{-6}/H_0$. The Model D outputs (spacing 0.1) sample the profile’s fast modes coarsely; the sums and E_h are exact at each output. The Model C quartic cross-check at 4×10^{-7} is CVODE’s accumulated phase error at the default tolerance, shown by the tighter run.

12 Reproducing this work

From a clone of the repository, with Rust (**cargo**), Python 3.10 or newer, **git** and a $\text{\LaTeX}$ distribution with **latexmk** installed:

```
bash fable-cosmology/setup.sh      # Linux, macOS, Git Bash on Windows
bash fable-cosmology/run_all.sh
```

or, in Windows PowerShell,

```
powershell -ExecutionPolicy Bypass -File fable-cosmology\setup.ps1
powershell -ExecutionPolicy Bypass -File fable-cosmology\run_all.ps1
```

The setup creates a Python virtual environment, makes a sparse clone of the platform’s **rustSolveIt** engine into **fable-cosmology/rust/vendor/rustSolveIt** and builds the solver. The second script runs the solver’s seven unit tests, executes the four notebooks in place (rewriting every CSV and PNG under **results/**), checks the notebooks against their eight requirements, and compiles this paper. The Mathematica side is

```
wolframscript -file claud-fable/run_from_nb.wls # 172 Input cells
wolframscript -file fable-cosmology/reference/make_reference.wls # NDSolve CSVs
```

Every number of this paper is read from a file those commands write.

A CSV columns

scalar-flrw (Model A)

a, z, N, t, phi, phidot, H, KE, PE, rho_phi, P_phi, w_phi, Omega_phi,
Omega_m, Omega_r, q_dec

scalar-cpl (Model A')

a, z, N, t, phi, phidot, H, KE, PE, rho_phi, P_phi, w_phi, w_cpl,
Omega_phi_test, Omega_DE_cpl

spinor-flrw (Model B)

a, z, N, t, s, H, K_Psi, U_Psi, rho_Psi, P_Psi, w_Psi, Omega_Psi, Omega_m,
Omega_r, q_dec; here $K_\Psi = sV'$ and $U_\Psi = \rho_\Psi = V$

scalar-pre (Model C)

x4, phi, phidot, KE, PE, rho, P, w, w_avg, rho_drift

scalar-pre-x0 (Model D)

x4, phi_mid, KE_mid, PE_mid, GO_mid, rho_mid, P_mid, w_mid, E_sheet,
P_sheet, w_sheet, KE_frac, PE_frac, GO_frac, E_drift; with `--profile-out` a
second file whose header is x4 followed by one phi per grid point, whose first data row is NaN
followed by the x^0 grid, and whose following rows are x^4 and $\phi(x_j^0, x^4)$

Numbers are written in the Rust format `{:.15e}`: one digit, a point, fifteen decimals and an exponent, for example `9.118819655545162e-4`, i.e. 16 significant digits.

References

- [1] M. Chevallier and D. Polarski, “Accelerating universes with scaling dark matter”, Int. J. Mod. Phys. D **10** (2001) 213, arXiv:gr-qc/0009008.
- [2] E. V. Linder, “Exploring the expansion history of the universe”, Phys. Rev. Lett. **90** (2003) 091301, arXiv:astro-ph/0208512.
- [3] B. Ratra and P. J. E. Peebles, “Cosmological consequences of a rolling homogeneous scalar field”, Phys. Rev. D **37** (1988) 3406.
- [4] C. Wetterich, “Cosmology and the fate of dilatation symmetry”, Nucl. Phys. B **302** (1988) 668.
- [5] R. R. Caldwell, R. Dave and P. J. Steinhardt, “Cosmological imprint of an energy component with general equation of state”, Phys. Rev. Lett. **80** (1998) 1582.
- [6] R. R. Caldwell and E. V. Linder, “The limits of quintessence”, Phys. Rev. Lett. **95** (2005) 141301, arXiv:astro-ph/0505494.
- [7] R. R. Caldwell, “A phantom menace?”, Phys. Lett. B **545** (2002) 23.
- [8] A. Vikman, “Can dark energy evolve to the phantom?”, Phys. Rev. D **71** (2005) 023515.
- [9] B. Feng, X. Wang and X. Zhang, “Dark energy constraints from the cosmic age and supernova”, Phys. Lett. B **607** (2005) 35.
- [10] Y.-F. Cai and J. Wang, “Dark energy model with spinor matter and its quintom scenario”, Class. Quantum Grav. **25** (2008) 165014, arXiv:0806.3890.

- [11] E. J. Copeland, A. R. Liddle and D. Wands, “Exponential potentials and cosmological scaling solutions”, *Phys. Rev. D* **57** (1998) 4686.
- [12] J. A. Frieman, C. T. Hill, A. Stebbins and I. Waga, “Cosmology with ultralight pseudo Nambu–Goldstone bosons”, *Phys. Rev. Lett.* **75** (1995) 2077.
- [13] M. S. Turner, “Coherent scalar-field oscillations in an expanding universe”, *Phys. Rev. D* **28** (1983) 1243.
- [14] A. C. Hindmarsh, P. N. Brown, K. E. Grant, S. L. Lee, R. Serban, D. E. Shumaker and C. S. Woodward, “SUNDIALS: Suite of nonlinear and differential/algebraic equation solvers”, *ACM Trans. Math. Softw.* **31** (2005) 363.
- [15] D. J. Gardner, D. R. Reynolds, C. S. Woodward and C. J. Balos, “Enabling new flexibility in the SUNDIALS suite of nonlinear and differential/algebraic equation solvers”, *ACM Trans. Math. Softw.* **48** (2022) 31.
- [16] E. Hairer, S. P. Nørsett and G. Wanner, *Solving Ordinary Differential Equations I: Nonstiff Problems*, 2nd ed., Springer (1993).
- [17] P. Virtanen et al., “SciPy 1.0: fundamental algorithms for scientific computing in Python”, *Nature Methods* **17** (2020) 261.
- [18] rustSolveIt, Windows 11 edition with a pure-Rust SUNDIALS 7.8.0,
https://github.com/once-ere/rustSolveIt_Win11_SUNDIALS_7_8_0.
- [19] rustSolveIt, macOS Apple-silicon edition,
https://github.com/once-ere/rustSolveIt_macos-silicon_SUNDIALS_7_8_0.
- [20] rustSolveIt, Linux edition,
https://github.com/once-ere/rustSolveIt_linux_SUNDIALS_7_8_0.